

Oliver Donaldson

Data Science Student | Machine Learning & Data Engineering

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PROFILE

Third-year BSc Data Science student at Victoria University of Wellington. Industry experience automating an end-to-end operations workflow and building multi-layer Snowflake reporting over MongoDB and Kafka data at a FinTech lender. Currently shipping a production-grade ML platform that detects and ranks exoplanet transit candidates in NASA TESS and Kepler light curves.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, TypeScript, JavaScript

Machine Learning & AI: PyTorch, TensorFlow / Keras, scikit-learn, pandas, NumPy, dual-view 1D CNNs, MC-Dropout uncertainty, probability calibration, grouped k-fold cross-validation, Optuna, LLM optimisation

Statistics & Modelling: Statistical modelling (GLMs, shrinkage methods), clustering, A/B testing, statistical coding in R

Databases & Data Engineering: Snowflake, PostgreSQL, SQLite, MongoDB,, Kafka, ETL (Hevo), n8n, workflow automation, tf.data

MLOps & Tooling: Git, MLflow, DVC, Prefect, Hydra, Pandera, Docker, GitHub Actions, CI/CD, pytest, Ruff, mypy

Web & Visualisation: FastAPI, React, Vite, Tableau, Grafana, R Studio

Governance & Ethics: AI governance, data privacy, regulatory frameworks

Currently developing: ArcGIS and spatial analysis (GEOG 215)

PROFESSIONAL EXPERIENCE

AI & Analytics Intern | *Harmony — Financial Services*

Nov 2025 – Feb 2026

- **End-to-end feature automation:** built the "Obligation Met Letter" loan-closure confirmation, replacing a manual Operations process handling ~70–80 requests per month. Implemented an 8-day eligibility buffer, system-event logging for audit trails, and a self-service download dashboard for customers.
- **Data engineering:** developed a multi-layer reporting framework in Snowflake, building SQL views that transformed raw MongoDB and Kafka data into Tableau-ready datasets for Finance teams.
- **Real-time monitoring:** built "StatsBot" using n8n, Kafka and MongoDB aggregation pipelines to serve real-time API ground-truth statistics, replacing legacy reporting systems.
- **AI governance:** authored GitHub Copilot system instructions to standardise PR reviews around security patterns and architectural consistency.
- **Stack:** Snowflake (SQL), MongoDB, Kafka, n8n, Tableau, TypeScript, React, Python, SendGrid.

Sailing Coach | *Royal Akarana Yacht Club*

May 2019 – Jan 2024

- Led development programmes across a range of age groups, focusing on safety, confidence-building, skill development and leadership.
- Communicated complex technical skills to groups ranging from complete beginners through to competitive racing squads.

Catering Assistant | *Compass Group NZ / Air New Zealand Lounge*

Sep 2023 – Present

- Deliver premium hospitality service in a high-volume lounge environment while studying full-time, balancing a fixed weekend roster against academic deadlines.

PROJECTS

Exoplanet Hunter | *ML platform for transit detection in NASA TESS & Kepler data*

v2 in development

github.com/OliverDonaldson/exoplanet-hunter-v2 · [live vetting console](#)

Self-directed research project that classifies brightness dips in unreviewed light curves as genuine planetary transits or false positives, ranking candidates by calibrated probability so the most promising can be prioritised for human follow-up.

- Trained a dual-view (global + local) 1D CNN with attention on real MAST light curves only — no synthetic data in the training set — reaching parity with published baselines (ExoNet) under 5-fold grouped cross-validation.
- Built the full data path in Python: MAST TAP catalogue queries, light-curve cleaning, flattening and phase-folding, and a streaming tf.data input pipeline over versioned Parquet datasets.
- **v2 rebuild:** automated catalogue refresh, Pandera data-validation gates in CI, on-demand GPU-burst training, calibrated ensemble inference with MC-Dropout uncertainty estimates, and a FastAPI scoring endpoint.

- Built an interactive React + Vite console for vetting candidates, backed by MLflow run tracking, DVC dataset versioning and Prefect orchestration.
- Held every component to a Random Forest baseline before adopting it — nothing ships unless it beats the simplest working solution.
- **Stack:** Python, TensorFlow 2.16, scikit-learn, FastAPI, React 18, TypeScript, Hydra, MLflow, DVC, Prefect, Optuna.

Study Tracker | *Interactive macOS desktop widget*

- Built a glanceable desktop widget for managing university study — course-by-course progress, daily task lists and user-managed courses, editable in place without opening a separate app.
- Rendered the UI in React JSX on the Übersicht widget engine, with custom SVG for a knowledge-tree visualisation that grows as tasks are completed.
- Implemented local persistence by serialising state to JSON on every change, so courses, edits and completions survive refreshes and restarts.
- Styled with CSS-in-JS and custom animations for plant growth and an XP/levelling progression.
- **Stack:** React (JSX), Übersicht, SVG, CSS-in-JS, JavaScript.

F.R.I.D.A.Y. | *Finance, Resources, Information, Data, Assistance and Yield*

CS50x final project, 2021

- Financial guidance web app built as the capstone for Harvard CS50x, helping users improve savings habits and reach practical advice quickly.
- Built a voice assistant for hands-free navigation of the site's tools and guidance content.
- Designed for phone and desktop with a collapsible navbar, responsive page sizing and accessible navigation.

EDUCATION

BSc Data Science | *Victoria University of Wellington*

2024 – Expected 2028

- **Relevant coursework:** DATA 305 Machine Learning Techniques, DATA 303 Statistics for Data Science, AIML 231 Techniques in Machine Learning, AIML 232 Techniques in AI, DATA 202 Data Management & Programming, DATA 101 Introduction to Data Science, STAT 292 Applied Statistics, STAT 193 Statistics in Practice.
- **In progress (2026):** DATA 301 Data Science in Practice, STAT 293 Applied Statistical Modelling, GEOG 215 Introduction to GIS & Science, COMP 103 Data Structures & Algorithms.

Law Coursework | *Victoria University of Wellington*

2024 – 2025

- Completed LAWS 121 Introduction to the NZ Legal System, LAWS 122 Introduction to Case Law, LAWS 123 Introduction to Statute Law and LAWS 297 Introduction to Legal Skills as part of a BSc/LLB conjoint, before specialising in Data Science.

NCEA Level 3 (Excellence) | *St Peter's College, Epsom*

2017 – 2023

LEADERSHIP & ACTIVITIES

General Committee Member | *VicAI — Victoria Artificial Intelligence Society, VUW*

Jun 2026 – Present

- Organising a student Project Expo: lightning talks from postgraduate students across data science, AI and cyber security, aimed at encouraging undergraduates to build their own portfolio projects.
- Ran volunteer rostering and member communications for the club stand at the university Clubs Expo.

Competing Athlete | *NZIFBB*

2025

- Competed in natural bodybuilding, requiring sustained discipline and long-horizon goal-setting alongside full-time study.

Youth Training Programme | *Royal New Zealand Yacht Squadron*

2022

- Selected for the Mastercard Youth Training Programme; reached the quarterfinals in match racing.

CERTIFICATIONS

- **CS50x: Introduction to Computer Science** — Harvard University, 2021.
- **Sail Coach Certification** — Yachting New Zealand, 2017.